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Walking motion real-time detection method based on walking stick, IoT, COPOD and improved LightGBM

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Abstract

Real-time walking behavior monitoring is essential in ensuring safety and improving people's physical conditions with mobility difficulties. In this paper, a real-time walking motion detection system based on the intelligent walking stick, mobile phone and multi-label imbalance classification method combining focal loss and LightGBM (MFGBoost) is proposed. The Internet of Things (IoT) technology is utilized for communicating between the walking stick and mobile phone. The new MFGBoost is embedded into the Raspberry Pi to classify human motions. MFGBoost is scalable, and other boosting models, such as XGBoost, could also be used as its base classifier. An improved derivation method of the multi-classification focal loss function is proposed in this paper, which is the key to the combination of multi-classification focal loss and Boosting algorithms. We propose a novel denoise method based on window matrix and COPOD algorithm (W-OD). The window matrix is designed to extract data features and smooth noise, and COPOD could output the noise level of the model. A weighted loss function is designed to adjust the model's attention to different samples based on the W-OD algorithm. We evaluate the latest classification model from multiple perspectives on multiple benchmark datasets and demonstrate that MFGBoost and W-OD-MFGBoost could improve classification performance and decision-making efficiency. Experiments conducted on human motion datasets show that W-OD-MFGBoost could achieve more than 97 percent classification accuracy.

Keywords Walking stick motion detection · IoT · Raspberry Pi · MFGBoost · W-OD algorithm

1 Introduction

Human motion detection plays a vital role in medical care, motion analysis and disease detection. The traditional motion detection methods based on strain sensors have relatively single functions and weak expansibility. The intelligent walking stick is a kind of auxiliary movement equipment with both supporting action and algorithm analysis. It has the characteristics of convenience, simple operation and rich functions. However, the current researches on intelligent walking stick mainly focus on fall detection and alarm, and there is few detailed analysis of movement behavior.

In this paper, we propose a real-time walking motion detection system based on the intelligent walking stick, mobile phone and MFGBoost. The framework of the intelligent walking stick is shown in Fig. 1. The acceleration and gyroscope information of the walking stick is collected by the upper and lower MPU-6050. The machine learning algorithm is embedded into Raspberry Pi to realize real-time motion classification. The IoT system based on the walking stick, mobile phone application (APP) and cloud database is built to realize the real-time remote transmission of information between the walking stick and APP.

Since classification models need to be migrated to small embedded devices, we mainly discuss low-power machine learning models, such as spiking neuron networks (SNN) and ensemble learning algorithms. SNN could realize the decision-making based on neuromorphic computing [1], which has lower power consumption and shorter computing time compared with the traditional neural network. In [2],

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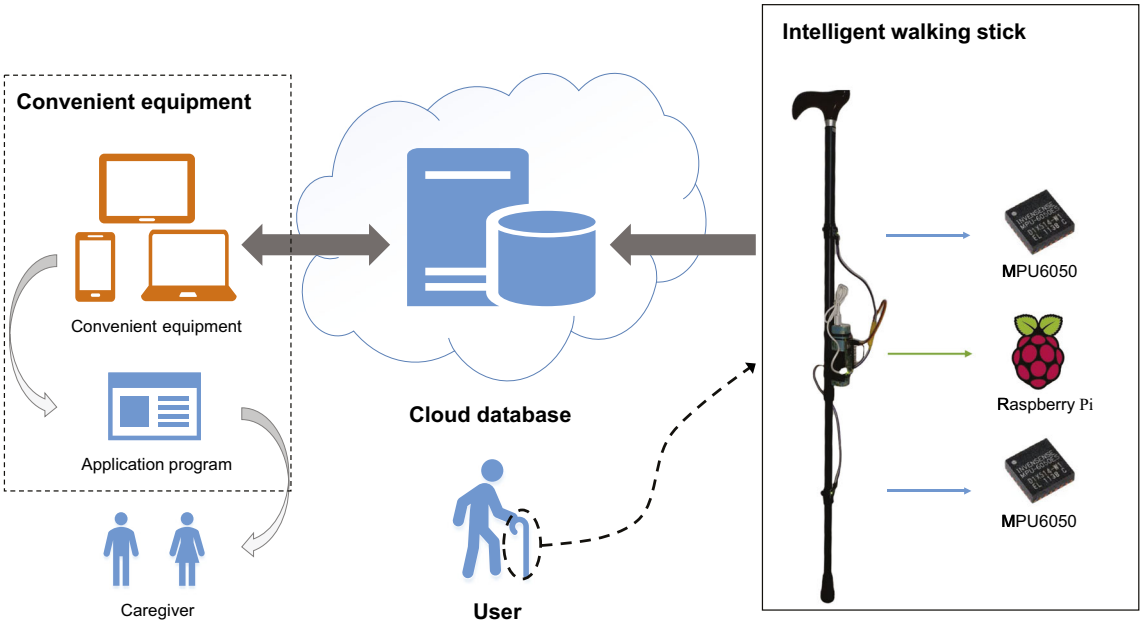


Fig. 1 Intelligent walking stick system based on IoT, MPU-6050 and Raspberry Pi

SNN was designed to control the LEGO robot, which could exhibit classical and operant conditioning. However, the function of this algorithm was relatively simple and could not meet the requirements of the walking stick motion analysis. SNN was improved to achieve more efficient brain-like computing in [1, 3]. These methods work best in digital and mixed-signal neuromorphic hardware but are not suitable for ordinary embedded devices. SNN has more significant computing potential than the ensemble learning algorithm but fails to achieve better computational accuracy. As an important part of machine learning, ensemble learning completes the task by building multiple learners. As one of ensemble learning algorithms, boosting learning algorithm is better than artificial neural networks in analyzing small and medium-sized data, such as sensors data. Algorithms, such as LightGBM [4] and XGBoost [5], are far ahead of other ensemble learning algorithms regarding classification efficiency and accuracy. Some of them were applied in many fields and achieved good results [6–8].

Although the boosting algorithm has succeeded in the classification problem, it fails to overcome the class imbalance (label-imbalance) problem. Classification under label-imbalance situations commonly exists in practice. Most machine learning methods suffer performance decline if the ratio among labels or the convergence difficulty of different category's samples differs significantly [11]. For example, in the process of human motion, the frequency of walking motions is significantly higher than that of other motions. The difficulty of collecting different types of data is different, which could result in class imbalance. In addition, since the learning difficulty of different motions varies, it leads to class imbalance. In [9–13], the influences of class imbalance on boosting and other ensemble learning models were investigated.

Most papers on class imbalance problem mainly focus on the binary label-imbalance classification problem, and it is rare to apply the improved boosting algorithm to solve the multi-label class imbalance problem. Making use of the efficient learning ability of the boosting algorithm to overcome class imbalance are significant.

In this paper, the focal loss function [14] is improved to solve the multi-label imbalance classification problem of boosting model. The difficulty of focal loss in finding the second derivative is solved by replacing the first derivative with $q_e f_1' + (1 - q_e) f_2'$. f_1' and f_2' correspond to two kinds of outputs of Softmax derivative respectively, and q_e is the true value. The improved focal loss could combine with boosting model, such as LightGBM and XGBoost, to constitute the MFGBoost. MFGBoost could solve the multi-classification class imbalance, and it has higher decision-making efficiency.

Due to the characteristics of the sensor, collecting data often brings some noise. In addition, the diversity of cane movement decreases the signal-to-noise ratio of data collected by the sensors in the walking stick. Data noise reduces the accuracy of classification. However, the noise problem is rarely investigated in the existing motion analysis.

In this paper, we aim to use the denoise algorithm to reduce the impact of noise on the accuracy of classification. Most of the denoise algorithms analyze the data directly and

remove the noisy information. However, the classification model trained with incomplete data could cause lower accuracy when classifying in real-time. Unlike the traditional denoise algorithm, the outlier detection algorithm could be combined with the ensemble learning algorithm to classify noisy data without losing complete information. COPOD [15] is an outlier detection algorithm based on copulas, which has a remarkable effect on the detection of outliers in multi-dimensional data. In this paper, a novel denoise method W-OD based on window matrix and COPOD is proposed. By increasing the model's attention to the samples with outliers, the learning ability of the model to the data with a low signal-to-noise ratio is improved. The window matrix is the minimum unit in data analysis. COPOD algorithm detects each data's outlier degree, which indicates the extent of the data polluted by noise. We mark the maximum outlier degree value of data in the window matrix and adjust the weight of the loss function on this window matrix according to this outlier degree value. Therefore, W-OD realizes the greater attention to noisy data without losing information. Through W-OD algorithm and the robust feature learning ability of the boosting learning algorithm, the models could obtain good classification results even if there is a noise influence on the sample data.

The conventional boosting algorithms, such as CATBoost [16] and XGBoost, were poorly suitable for portable embedded devices because of their complex model, large amounts of calculation, long operation time, and less precision. LightGBM was proposed in 2017 to reduce data memory, achieve high accuracy, and ensure that a single machine could use much sample data without lowering speed. Therefore, we build W-OD-MFGBoost based on LightGBM and transplant it to the intelligent walking stick for motion analysis.

Experimental analysis of MFGBoost and W-OD algorithms on multiple benchmark datasets shows that our algorithms have better performance.

The contributions of this paper are summarized as follows.

(1) We propose a walking motion real-time detection method based on walking stick, IoT, COPOD and improved LightGBM, which could realize motion behavior detection, real-time motion information transmission and motion state analysis.
(2) We propose a novel multi-label imbalance classification method combining focal loss and LightGBM named MFGboost to solve the class imbalance when using a boosting algorithm to deal with multi-classification tasks.
(3) We propose a novel denoise method based on window matrix and outlier detection (W-OD), which reduces the impact of noise on classification results without changing the sample data.

The rest of this paper is arranged as follows. Section 2 introduces the related work. Section 3 presents the composition method of the walking stick system based on IoT. Section 4 elaborates the proposed MFGBoost and W-OD algorithm. Section 5 introduces the datasets and shows the results of extensive ablation experiments and comparative experiments. Section 6 summarizes this paper.

2 Related work

2.1 Human motion analysis of the intelligent walking stick

The motion detection of the human body was investigated in a large number of papers [17–20]. In [17], the method of detecting and labeling reliably biological motion was proposed. The different types of human motions, including typing, breathing and speaking, were detected based on the carbon tube sensors in [18]. After that, several types of strain sensor were designed and utilized in human motion detection in [19, 20]. Since the strain sensor could measure the strain of human activities, it has received great attention in recent years. However, these strain sensors were affixed in the human body, which was not convenient for human activities.

Compared with wearable motion analysis equipment, the intelligent walking stick is suitable for people with mobility difficulties. To help those people to walk safely and independently, auxiliary intelligent walking sticks with sensors and algorithms were proposed in [21–24]. In [21], an intelligent walking stick was designed, which was equipped with a multi-sensor system to detect obstacles. A robotic cane was proposed in [22], which could help users to stand and maintain their balance while standing and walking. In [23, 24], the intelligent sticks on admittance control were designed to provide safe and efficient gait training for the elderly. However, the above researches failed to consider the communication between caregivers and walking stick users, and the motion information could not be transmitted to the caregivers in real-time to achieve better nursing results.

With the development of society, the communication between intelligent walking sticks and people has become increasingly important. Traditional communication methods failed to meet the needs of people for long-distance real-time communication. With the development of the cloud, more intelligent devices have overcome this problem through IoT. In [25–27], IoT technology was studied to

achieve remote monitoring by intelligent devices. However, there were few applications of IoT in smart walking sticks.

2.2 Imbalanced data classification

Class imbalance is a common problem in machine learning [28]. When applying machine learning techniques in human motion studies, imbalance in class distribution may influence the performance of a classifier [29]. The data resampling, either oversampling the minority class instances or undersampling the majority class instances, is the most common approach [30–33]. However, this method has high computational complexity. In [30], a clustering-based undersampling technique was proposed to address the class imbalance, which balanced the predictive performance and the complexity. It only discussed the problem of binary-classification but did not consider the case of multi-classification. Furthermore, these methods pay too much attention to the data and ignore the classifier's performance. The improved boosting algorithms were adopted to overcome the class imbalance [34, 35]. However, these models still have a long classification time and low classification accuracy compared with LightGBM.

Focal loss is an improved cross-entropy loss function, which introduces a $(1 - p_j)^{\gamma}$ factor to the cross-entropy function (where p_j is the prediction of $p(x = j)$). Focal loss solves class imbalance without decreasing the overall classification accuracy of the classifier. It enhances the model's attention to the samples with low classification accuracy and reduces the attention to those with higher classification accuracy in the training process. The focal loss was utilized in target detection, natural language processing, image segmentation and other aspects [36–38]. Due to the unique property, the focal loss could also work with some ensemble learning algorithms to adjust the sample weight and improve the model [9, 10, 13]. XGBoost combining with focal loss was built for vehicle re-identification in [9]. In [10], the sample imbalance during the training stage of Adaboost was solved by the focal loss function. The focal loss function was utilized in LightGBM to predict potential solar adoption in [13]. These results focus on the binary classification of focal loss function, and there are few results on focal loss in the multi-classification ensemble model. In [39], OVR (One Vs Rest) algorithm and sigmoid function were utilized to transform the focal loss-based LightGBM binary classification into the multi-classification, which would cause a lot of time and space consumption. It is still a challenge to adopt focal loss and Softmax function to solve the multi-classification problem. The multi-classification focal loss problem under the Softmax activation function is not addressed in the boosting learning model, such as LightGBM and XGBoost. The solution of multi-classification focal loss would encounter difficulty in finding the second derivative, which is caused by two kinds of outputs of Softmax derivative.

2.3 Denoise of the data collected by sensors

When the sensors in the moving device are utilized for detection, the collected data often have a certain degree of noise. Many papers on intelligent walking stick failed to discuss this issue [22–24]. In [21], a dynamic feedback compensator was designed as a noise filter. However, it is difficult to use this method to deal with noise interference in high-dimensional data classification.

Many studies on the denoise algorithm of data were proposed in [40–42]. The general bayesian wavelet packet method based on the minimum mean square error was developed in [40]. In [41], a novel multivariate signal denoising method was presented to compute the Mahalanobis distance measure at multiple data scales. A signal denoising approach based on a specialized ResNet was proposed in [42]. These methods could denoise the data before modeling and change the value of noisy data directly. However, they could reduce the complexity of the dataset, and the training process is likely to lead to over-fitting.

3 Intelligent walking stick system

3.1 Walking stick system composition

The intelligent walking stick system adopts two MPU-6050 processors to collect 3-axis acceleration and 3-axis gyroscope information, which are located in the walking stick's top and bottom, respectively. This arrangement enables the sensors to collect more comprehensive trajectory information of the walking stick. Raspberry Pi zero w is located in the middle, a special edition released by the Raspberry Pi Foundation in 2017. It has a 1 GHz single-core CPU, 512 MB memory, and a 40-pin GPIO expansion port. Most machine learning algorithms, such as XGBooist and LightGBM, could be embedded into Raspberry Pi. Raspberry Pi zero w has a small and exquisite appearance, which is suitably installed in portable devices. It supports a Wi-Fi module, which is essential for building wireless IoT. Figure 2 shows the composition and the network connection of the walking stick.

The Raspberry Pi connects the MPU-6050 through the jump cables, and a small mobile power supply is installed in the walking stick to support the Raspberry Pi. To ensure the internet connection of Raspberry Pi, we install portable Wi-Fi on the walking stick, which could provide a network anywhere.

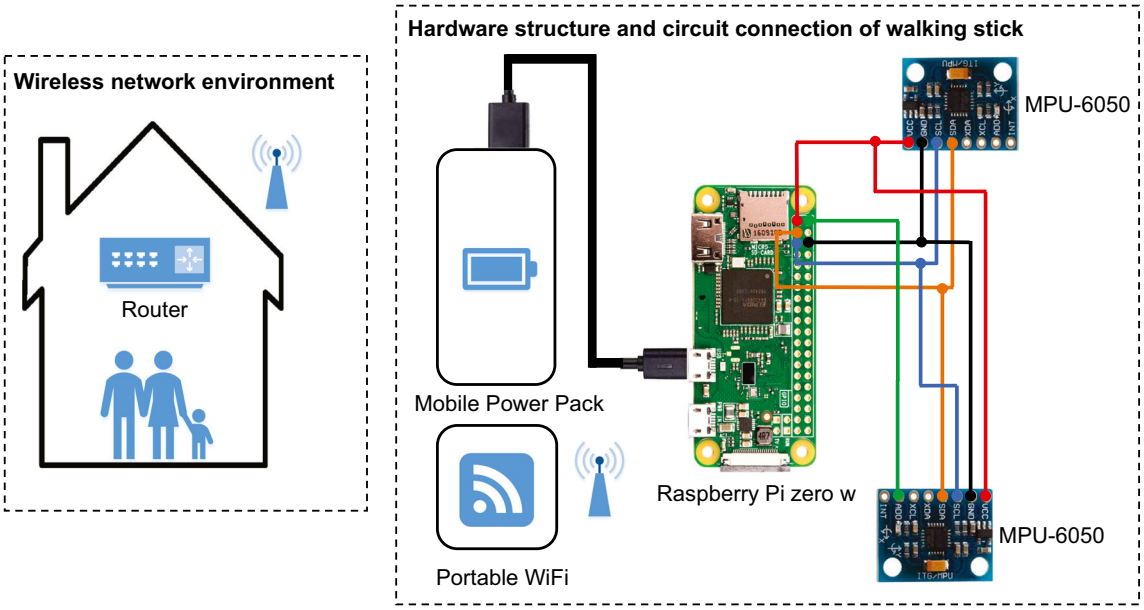


Fig. 2 The composition structure and network connection mode of a walking stick

3.2 IoT environment construction

We choose a relational database as the cloud database and compress the transmitted data into character types. The database is shown in Fig. 3.

The Hypertext Transfer Protocol (HTTP) is utilized to realize the data transmission from Raspberry Pi to the cloud database. It is a Client-Server (or Master-Slave) protocol. The Uniform Resource Locator (URL) information of the database is recorded, and the POST method is adopted to send the information to the database. The above operations are achieved through the Request library in the Python language.

The event-triggered communication mechanism is created by adjusting algorithm logic, which could improve the running speed of the model. When the motion state changes, the Raspberry Pi uploads the current motion state and records the duration time of the previous state. Raspberry Pi uploads duration for all motion states every minute. The specific operating logic is shown in Fig. 4. The relational database is adopted to store the uploaded information in the cloud, including the duration of each category of the motion, the current state of the motion and the date. The real-time state is expressed in numbers to facilitate data transmission.

The APP is designed to compose IoT system with the walking stick to monitor and analyze motion behavior. The GET and POST request methods are adopted in the mobile phone to read and modify the database information, including the current and the historical motion information. The operation logic of the mobile APP is shown in Fig. 5. Since the dictionary's typical variables, including duration

label: value: Query label: value: Add/modify

id	label	Value (double click to modify)	time
7094	otherstime2021-02-24	19.0	2021-02-24 21:11:00
7093	lefttime2021-02-24	6.0	2021-02-24 21:10:59
7092	righttime2021-02-24	8.0	2021-02-24 21:10:58
7091	fasttime2021-02-24	43.0	2021-02-24 21:10:58
7090	slowtime2021-02-24	38.0	2021-02-24 21:10:58
7089	stoptime2021-02-24	161.0	2021-02-24 21:10:57
8	state	1	2021-02-24 21:11:12

Fig. 3 The data stored in the relational database

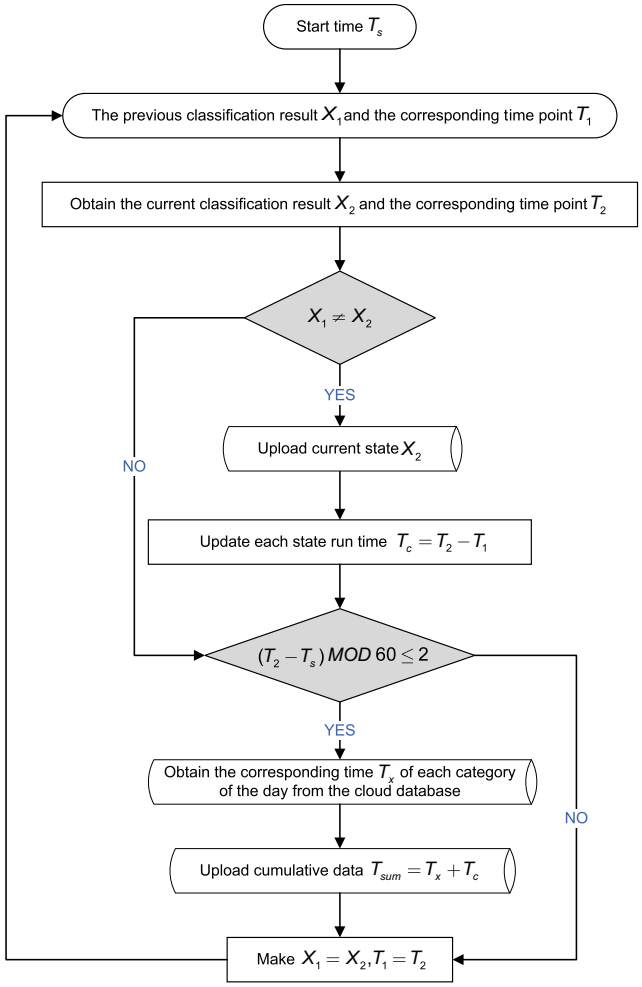


Fig. 4 Uploading logic of Raspberry Pi data

and motion state, are recorded in the relational database, the summary of the motion in a period could be obtained through the mobile phone APP. Records of the motion duration time for any day in history could also be queried. The monitoring and health analysis of the motion state can be realized through the motion analysis Application Programming Interface (API). The relevant interface of the mobile phone APP is shown in Fig. 6. The supervised motion duration refers to the motion duration after starting APP, which is realized by the logic of APP.

4 Improved gradient boosting algorithm

In this paper, the boosting model is improved from two aspects. MFGBoost algorithm is proposed to improve the classification performance of imbalanced data, and W-OD algorithm is designed to reduce the impact of the noise. These two algorithms are general extensions of boosting models and works with boosting models to achieve better classification results. Moreover, MFGBoost and W-OD could work together to form a more robust classification model W-OD-MFGBoost.

4.1 Multi-label imbalance classification method combining focal loss and LightGBM

4.1.1 Gradient boosting decision tree

The most widely used boosting algorithms, such as LightGBM and XGBoost, are constructed based on the gradient boosting decision tree (GBDT).

As an ensemble learning framework, GBDT is constructed from the linear combination of M weak classifiers with low classification accuracy.

For dataset $\{x_i, y_i\}_1^n$, $x = \{x_1, \ldots x_n\}$ is the input, and $y = \{y_1, \ldots y_n\}$ is the output.

$$F_M(x) = \sum_{m=0}^{M} \beta_m f_m(x) = \sum_{m=0}^{M} \beta_m h\left(x; a_m\right). \tag{1}$$

formula (1) is the GBDT model, which represents the mapping relationship from x to y. β_m is the learning rate, which denotes the step-size to further reduce the loss. $f_m(x) = h(x; a_m)$ is the m_{th} decision tree model, where α_m represents the best classifier parameters with the ability to fit the negative gradient. GBDT could solve the parameters by the iterative method. Defining the initial value of $F(x)$ as $F_0(x)$, its m_{th} iteration function is shown in formula (2).

$$F_m(x) = F_{m-1}(x) + \beta_m h\left(x; a_m\right). \tag{2}$$

Thus, the objective function of the m_{th} iteration is as follows:

$$\begin{aligned} Obj^{(m)} &= \sum_{i=1}^{n} L\left(y_i, F_m(x_i)\right) \\ &= \sum_{i=1}^{n} L\left(y_i, F_{m-1}(x_i) + \beta_m h\left(x_i; a_m\right)\right). \end{aligned} \tag{3}$$

Through the gradient descent algorithm, the negative gradient is adopted to update α_m and β_m constantly, and the decision tree model of m_{th} round could be obtained. As shown in Fig. 7, the final classifier is obtained through iteration and M decision trees combination.

4.1.2 Improved boosting algorithm MFGBoost

MFGBoost is a general extension of boosting algorithms, such as XGBoost and LightGBM. It provides multi-label imbalance data processing by combining these boosting models and improved focal loss function. The second-order derivative of the loss function is necessary for training the

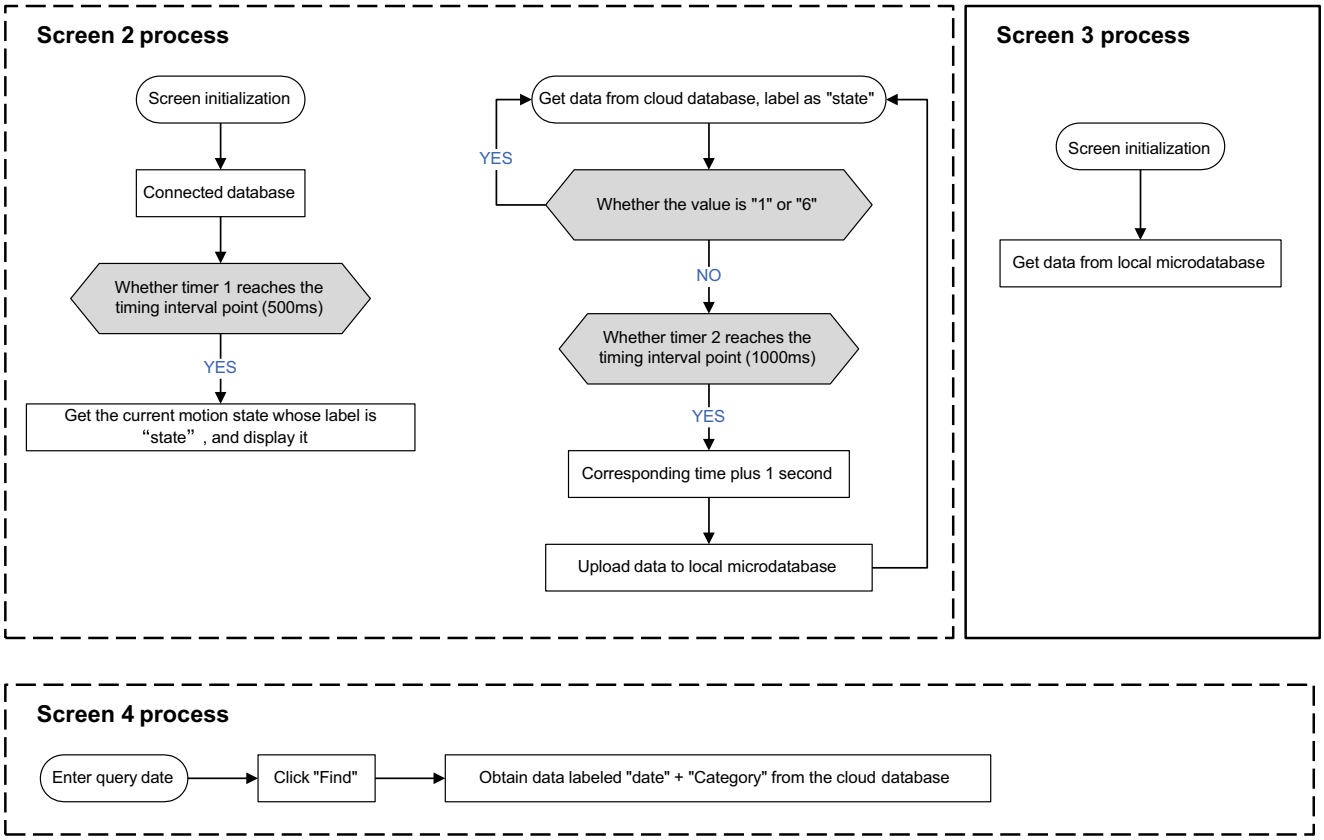


Fig. 5 Operating logic of mobile APP

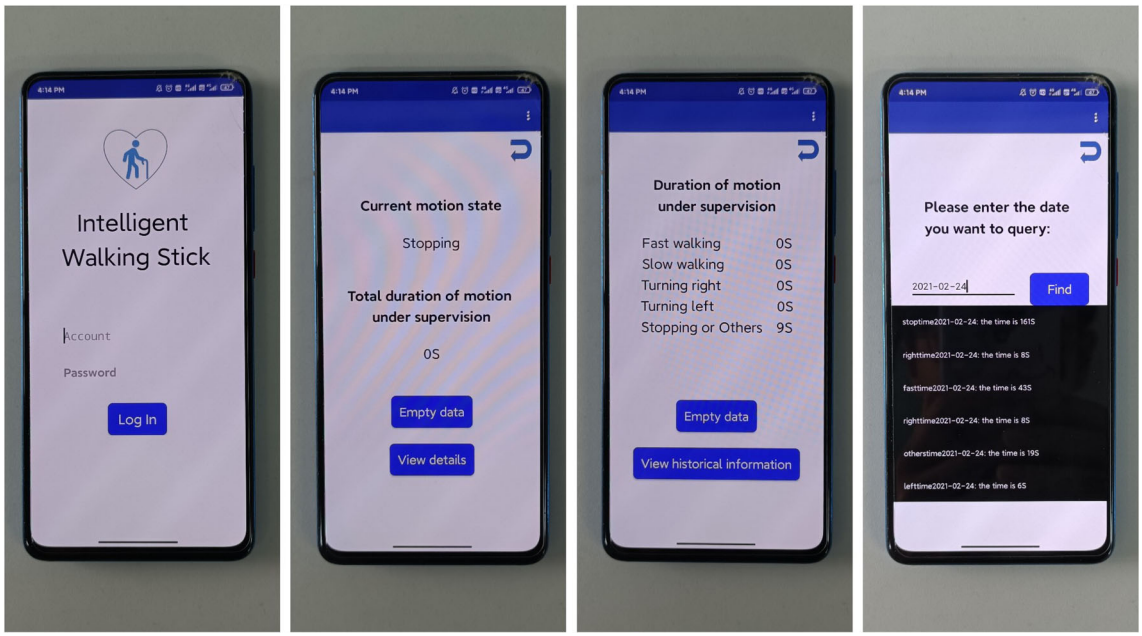


Fig. 6 The running interface of mobile phone APP

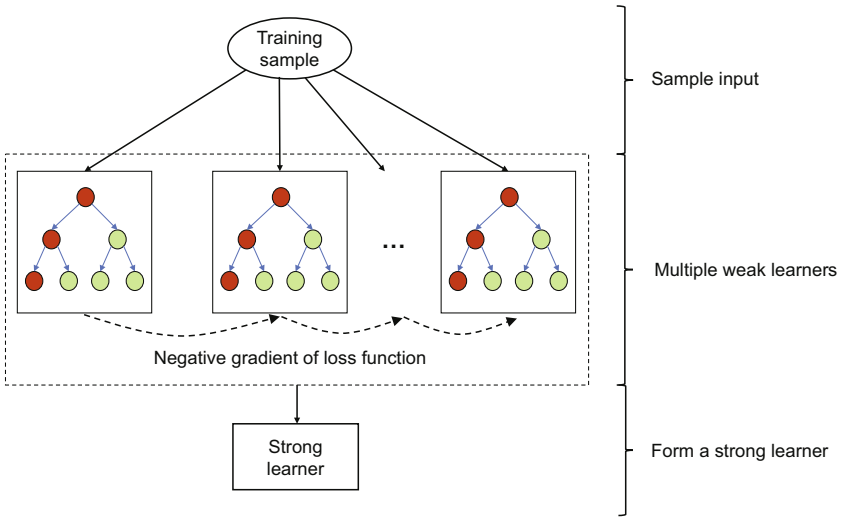


Fig. 7 GBDT model learning framework

model based on XGBoost, which increases the difficulty of solving multi-classification focal loss.

The objective function of algorithms based on XGBoost is shown in formula (4).

$$Obj^{(m)} = \sum_{i=1}^{n} L\left(y_i, \hat{y}_i^{m-1} + f_m\left(x_i\right)\right), \tag{4}$$

where $f_m\left(x_i\right)$ is the predicted value of the m_{th} tree for the input x_i, and $L\left(y_i, \hat{y}_i^{m-1} + f_m\left(x_i\right)\right)$ describes the loss between the predicted value $(\hat{y}_i^{m-1} + f_m\left(x_i\right))$ of the boosting model composed of the previous m trees and the true value y_i.

The models are simplified by Taylor expansion of the objective function, and the first and second derivatives of the objective function are utilized as the basis of node split.

When the Taylor expansion of the above formula is carried out, formula (5) is obtained.

$$Obj^{(m)} \approx \sum_{i=1}^{n}\left[L\left(y_i, \hat{y}_i^{m-1}\right) + g_i f_m\left(x_i\right) + \frac{1}{2} h_i f_m^2\left(x_i\right)\right], \tag{5}$$

where

$$g_i = \frac{\partial L\left(y_i, \hat{y}_i^{m-1}\right)}{\partial \hat{y}_i^{m-1}}, h_i = \frac{\partial^2 L\left(y_i, \hat{y}_i^{m-1}\right)}{\partial \hat{y}_i^{m-1}}. \tag{6}$$

$L\left(y_i, \hat{y}_i^{m-1}\right)$ is the loss function, which represents the loss between the predicted value $\hat{y}_i^{m-1}$ of the model composed of the previous $m-1$ trees and the real value y_i.

The Softmax function is utilized to normalize the output of the model, which maps the values to a real number between 0 and 1. The output of the model is processed by the Softmax function as follows:

$$p_j = \frac{e^{a_j}}{\sum_{s=1}^{K} e^{a_s}}, \tag{7}$$

where a_s is the value of the s_{th} position of the output vector, and p_j is the probability of the output category j. The predicted output $\hat{y}_i^{m-1}$ is formed by p_j $(j \in \{1, \ldots, K\})$.

The traditional cross-entropy loss function is adopted to solve the multi-classification problem, which is given as follows:

$$L\left(y_i, \hat{y}_i^{m-1}\right) = -\sum_{j=1}^{K} q_j \log\left(p_j\right). \tag{8}$$

In formula (8), y_i and $\hat{y}_i^{m-1}$ are $1 \times K$ dimensional vectors, and the K is the number of categories. The j_{th} elements of them are q_j and p_j, respectively. The value of q_j is 0 or 1. The p_j is obtained by formula (7), which represents the probability of the output category j.

To construct MFGBoost algorithm, the focal loss function is utilized to replace the conventional loss function of boosting algorithm. The regular focal loss function is as follows:

$$\begin{aligned} L_{focal\ loss} &= -\alpha \sum_{j=1}^{K} q_j (1-p_j)^{\gamma} \log\left(p_j\right) \\ &= -\alpha (1-p_t)^{\gamma} \log\left(p_t\right), \end{aligned} \tag{9}$$

where α and γ are constants. α balances the importance of the samples, and γ adjusts the speed of weights reduction of samples. $p_t = \hat{y}_i^{m-1} y_i^T$, is the term of the $\hat{y}_i^{m-1}$ corresponding to the term of y_i equaling to 1.

The first and second derivatives of $L_{\text{focal loss}}$ are obtained as follows:

$$\begin{aligned} g_i &= \frac{\partial L_{focal\ loss}}{\partial \hat{y}_i^{m-1}} \\ &= \left[\frac{\partial L_{focal\ loss}}{\partial a_1}, \frac{\partial L_{focal\ loss}}{\partial a_2}, \ldots, \frac{\partial L_{focal\ loss}}{\partial a_K}\right], \\ h_i &= \frac{\partial^2 L_{focal\ loss}}{\partial \hat{y}_i^{m-1}} \\ &= \left[\frac{\partial^2 L_{focal\ loss}}{\partial a_1}, \frac{\partial^2 L_{focal\ loss}}{\partial a_2}, \ldots, \frac{\partial^2 L_{focal\ loss}}{\partial a_K}\right]. \end{aligned} \tag{10}$$

The solution of the first and second derivatives of e_{th} term ($e \in \{1, \ldots, K\}$) in $\hat{y}_i^{m-1}$ is given as follows:

$$\begin{aligned} \frac{\partial L_{focal\ loss}}{\partial a_e} &= f' = \frac{\partial(-\alpha(1-p_t)^\gamma \log(p_t))}{\partial a_e} \\ &= \begin{cases} f_1', & e = t, \\ f_2', & e \neq t. \end{cases} \end{aligned} \tag{11}$$

The derivative of Softmax function is as follows:

$$\frac{\partial p_t}{\partial a_e} = \begin{cases} p_e(1-p_e), & e = t, \\ -p_e p_t, & e \neq t. \end{cases} \tag{12}$$

Through formulas (11) and (12), the following formula can be obtained:

$$\begin{aligned} f_1' &= \alpha\gamma(1-p_e)^\gamma p_e log(p_e) - (1-p_e)^{\gamma+1} \\ &= \alpha(-\gamma(1-p_e)^{\gamma-1} log(p_e) p_e \\ &\quad +(1-p_e)^\gamma)(p_e - 1), \end{aligned} \tag{13}$$

and

$$\begin{aligned} f_2' &= \alpha\gamma(1-p_e)^{\gamma-1} p_e p_t log(p_t) + (1-p_t)^\gamma p_e \\ &= \alpha(-\gamma(1-p_t)^{\gamma-1} p_t log(p_t) + (1-p_t)^\gamma)(p_e). \end{aligned} \tag{14}$$

Since there are two forms for Softmax to derive, it is challenging to find the second derivative of $L_{focal\ loss}$. In order to facilitate the calculation, a function is designed to represent the first derivative as follows, which is completely equivalent to the loss function obtained from the case-by-case discussion.

$$\frac{\partial L_{focal\ loss}}{\partial a_e} = f' = q_e f_1' + (1-q_e) f_2', \tag{15}$$

where q_e is the item of true value corresponding to a_e, and formula (15) is equivalent to formula (11).

The second derivative is as follows:

$$\begin{aligned} \frac{\partial^2 L_{focal\ loss}}{\partial a_e} &= \frac{\partial q_e f_1' + (1-q_e) f_2'}{\partial a_e} \\ &= \alpha(-2((p_e - 1/2) log(p_t) + p_e) \\ &\quad \gamma(-1+q_e) p_t (1-p_t)^{\gamma-1} \\ &\quad +{p_t}^2 p_e \gamma log(p_t) \\ &\quad \times(-1+q_e)(\gamma-1)(1-p_t)^{\gamma-2} \\ &\quad +q_e(\gamma log(p_e) + 2\gamma + 1)(1-p_e)^{1+\gamma} \\ &\quad +(-1+q_e)(p_e-1)(1-p_t)^\gamma \\ &\quad -log(p_e) p_e \gamma^2 q_e (1-p_e)^\gamma) p_e. \end{aligned} \tag{16}$$

Taking formulas (15) and (16) into formula (10), g_i and h_i can be obtained.

We build the MFGBoost model by the custom loss function of boosting algorithms, which is written by Python language. In the custom loss function which is built by formulas (6)-(16), the true value y_i and the predicted value $\hat{y}_i^{m-1}$ are input, and g_i and h_i functions are the output.

The codes of formulas (15) and (16) are programmed into machine language by jit decorator, which greatly improves the training speed of the model.

4.2 Denoise algorithm combining window matrix and COPOD

4.2.1 Window matrix

The window matrix has two main functions, extracting data features and "standard frame" in conjunction with COPOD to form W-OD algorithm.

For human motion dataset, the window matrix is utilized to extract the feature for expressive action. The walking data are collected by the walking stick to generate the human motion dataset. Each row of the dataset is a twelve-column vector. To express the information of an action cycle, we process the data by a window. Every n rows of samples are selected as a window, and the column elements are 12 acceleration and gyroscope values. n is the width of the window, and each window is a $n \times 12$ dimensional matrix. Ten statistical features are extracted from each column of window matrix, including median, mean, variance, skewness, kurtosis, minimum, maximum, etc. These features form a 1×120 row vector as a sample point. The processing method of the dataset is shown in Fig. 8.

The window matrix combined with COPOD could reflect the extent of noise influence. The method of detecting noise is shown in Fig. 9.

In COPOD, an empirical copula is built to predict tail probabilities of each given data point to determine its level

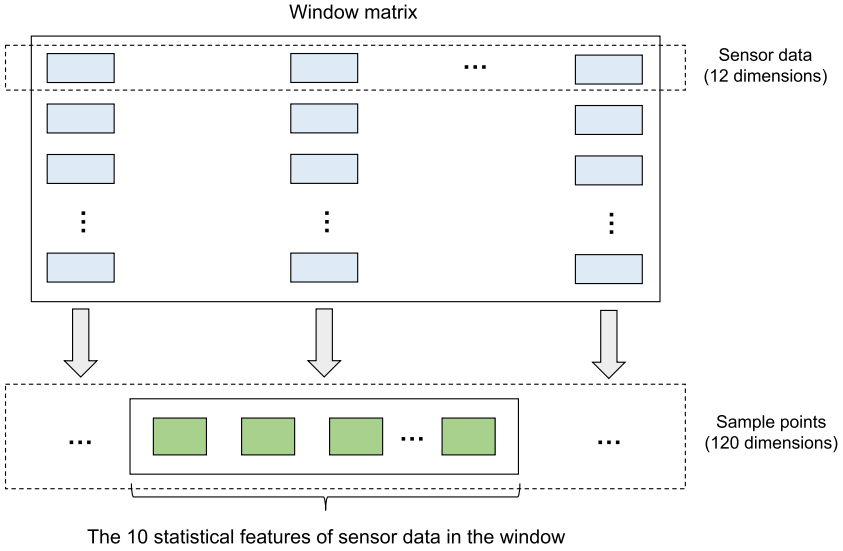


Fig. 8 The feature extraction of dataset

of "extremeness", which is an anomalous p-value [15]. In this paper, the anomalous value is regarded as the noisy degree of the sample point. The maximum anomalous value in this window matrix is taken as the noisy degree of this part sample.

4.2.2 The W-OD algorithm

The data collected by inertial sensors often contain some noise. Most of the existing algorithms preprocess the noisy data to reduce the impact of noise, which reduces the authenticity of the data. The traditional classification models use the same weight to learn all data during training, which reduces the classification accuracy of noisy data. To increase the classification accuracy of noisy data, it is necessary to improve the gradient of the noisy data, that is, to increase the attention of the model to noisy data. An outlier is an observation that lies an abnormal distance from other values in a random sample from a population. The samples affected by noise are often divorced from the global distribution of data, so they could be regarded as outliers. In this paper, W-OD algorithm obtains the degree of noise pollution by the outlier detection, and it makes the model pay extra attention to noisy samples by adjusting the weight of loss function to different samples. It enhances the classification performance of the model for noisy samples. W-OD could be used to adjust any form of the loss function, such as the cross-entropy loss function and its improved version.

The traditional cross-entropy loss function is improved by enhancing the classification gradient of noisy data, which is shown in formula (17).

$$L_{W-OD} = -\xi \sum_{j=1}^{K} q_j \log\left(p_j\right). \tag{17}$$

There are two values for ξ, which represent two weights adopted for different data. For noisy data, the larger value of ξ is ξ_1, which makes the model more potent to learn noisy data. For ordinary data, the smaller value ξ is ξ_2. COPOD algorithm is used to determine whether this part of the data belongs to noisy data.

For regular focal loss function in formula (9), the effect of α and γ is shown in Fig. 10. For α taking a smaller value

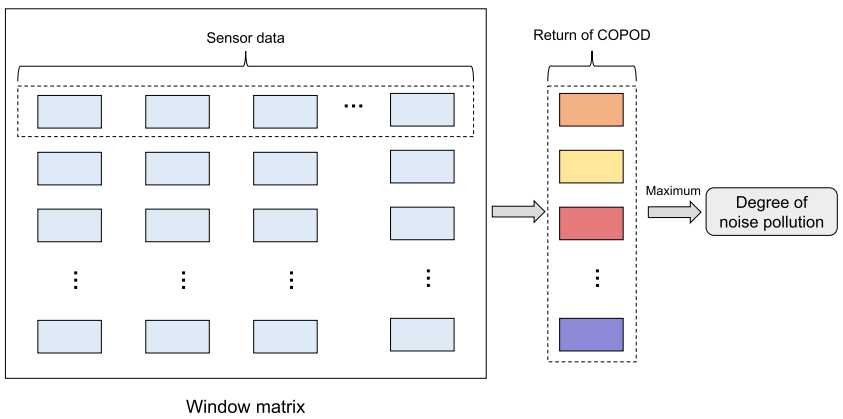


Fig. 9 Window matrix combines with COPOD to detect noise

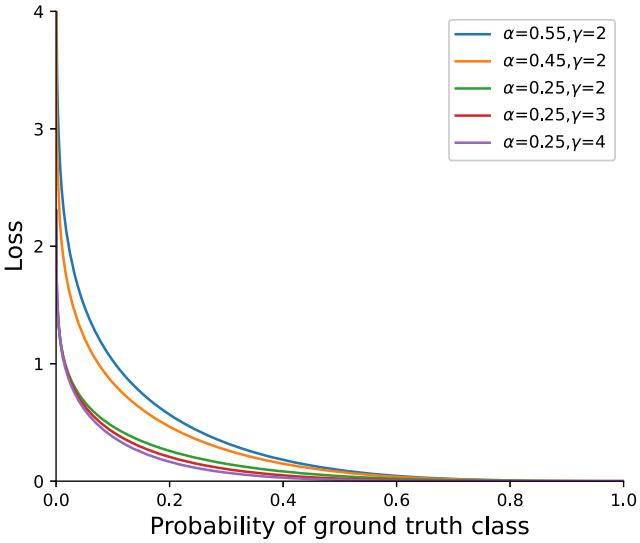


Fig. 10 The influence of α and γ on focal loss

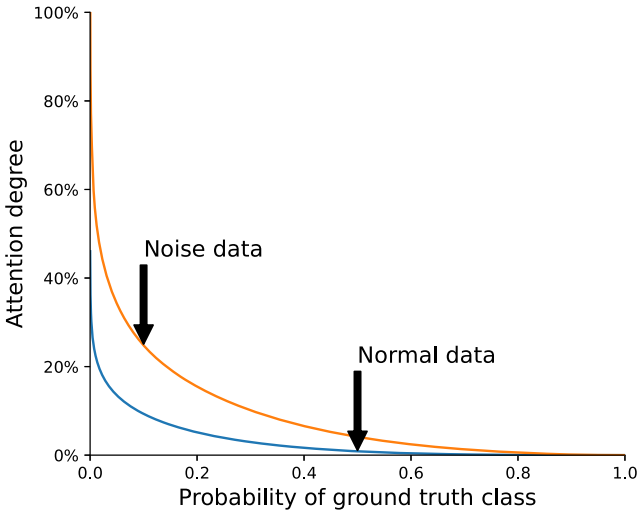


Fig. 11 Comparison of training intensity between normal samples and noise samples in W-OD

and γ taking a larger value, the samples with poor classification effect get a larger gradient, and the model reduces the attention to the samples with good classification effect. Taking a larger α and a smaller γ for some samples would increase the attention of the model to these samples.

The focal loss function is improved in formula (18).

$$L_{W-OD} = -\varepsilon(1-p_t)^{\beta}\log(p_t)\,. \tag{18}$$

The coefficients ε and β are designed by window matrix and COPOD, which are shown in formula (19) and (20).

$$\varepsilon = \begin{cases} \varepsilon_1, window_i > thr, \\ \varepsilon_2, window_i \le thr, \end{cases} \tag{19}$$

and

$$\beta = \begin{cases} \beta_1, window_i > thr, \\ \beta_2, window_i \le thr, \end{cases} \tag{20}$$

where the value of $window_i$ represents the influence degree of noise in the i_{th} window matrix, and thr is the threshold value to judge whether it is seriously affected by noise. For human motion dataset, the value of $window_i$ is expressed as formula (21).

$$window_i = \max_p \; COPOD(x_{pq}), \tag{21}$$

where $x_{pq}(p \in \{1,\ldots,n\}, q \in \{1,\ldots,12\})$ represents the value of the sample in the window matrix. The output of COPOD function is a $n \times 1$ dimensional vector, and its elements represent the noise outliers of the corresponding samples. The COPOD function is built by the anomaly detection toolbox PyOD [43].

When the samples affected by noise are given a larger ε and a lower β, the samples could obtain a larger gradient and attract more attention, which is shown in Fig. 11. Thus, W-OD enhances the denoise performance.

W-OD improves the focal loss algorithm of multi-label imbalance classification. Applying W-OD in focal loss forms a new W-OD-MFGBoost, it ensures the reliability of multi-label imbalance data classification and overcomes noise interference on data classification. W-OD-MFGBoost is finally transplanted to Raspberry Pi to classify the walking stick motion data.

5 Experiments

In this section, the contributions of this paper are analyzed from imbalance data classification and reducing noise interference. Three datasets published in UCI Machine Learning Repository [44] (MFCCs dataset, AReM dataset and Page Blocks dataset) and MINIST dataset [45] are adopted to show the superiority of MFBoost in class imbalance classification. The human motion dataset and HAPT dataset [46] are utilized to show the denoise effect of W-OD algorithm. Moreover, the ablation studies are conducted on HAPT dataset to analyze the individual components. We also conduct comparative experiments on HAR dataset [47] to show that our algorithm has a higher accuracy rate than other algorithms. Finally, the detailed experimental analysis of W-OD-MFGBoost is carried out on the human motion dataset to verify classification performance.

5.1 Dataset

5.1.1 Human motion dataset

The human motion dataset is built to train the classification model of the walking stick system, which is collected based on the walking stick with Raspberry Pi and MPU-6050.

Table 1 The details of imbalanced datasets

Dataset	Trains	Tests	Minority	Majority	Ratio
MFCCs	5036	2159	68	4420	1:65
Page Blocks	4925	548	28	4913	1:175
AReM	29567	12672	7200	2880	1:2.5
MINIST	55000	10000	4987	6179	1:1.2

The first three elements represent the x-axis, y-axis and z-axis accelerometers of the top MPU-6050. The second three elements represent the x-axis, y-axis and z-axis gyroscopes of the top MPU-6050, respectively. Similarly, the third and fourth groups elements represent the x-axis, y-axis, z-axis accelerometers, and x-axis, y-axis, z-axis gyroscopes of the bottom MPU-6050, respectively. The walking motions are divided into six categories, which are "fast walking", "slow walking", "turning left", "turning right", "stopping" and "others". The collected inertial sensor dataset contains 600000 samples. After the feature is extracted through window matrix (30 readings/window) of W-OD algorithm, each sample represents the motion information in an action cycle. Sixty percent of the dataset is utilized for training, twenty percent for validation, and the remaining twenty percent for testing.

5.1.2 HAR dataset and HAPT dataset

HAR dataset is a human activity dataset collected by smart phones worn at the people's waist, which consists of six categories of activity data. These data are obtained by filtering and sampling the sensor signal.

HAPT dataset is an upgraded version of HAR dataset, which includes labels of postural transitions between activities and the full raw inertial signals [44]. The original signal in HAPT dataset is processed to form the final version by the sampling and denoising methods introduced in [46]. In this paper, our window matrix could preprocess the original signals (60 readings/window), and W-OD algorithm reduces noise.

5.1.3 Imbalanced dataset

The four benchmark datasets are utilized to evaluate the classification performance of MFGBoost. The details of benchmark datasets are shown in Table 1.

Majority is the largest number of samples in one category, and Minority is the smallest number of samples in one category. The ratio represents the ratio of Minority to Majority. MFCCs and Page Blocks have great imbalance rates. The imbalance rates of MINIST dataset are low, but it is more difficult to classify some categories, such as "7" and "9". Similar to MINIST, this is also the case with AReM.

5.2 Detailed experimental analysis

The experiment is carried out on a laptop configured as 2 cores 4 threads Intel (R) Core (TM) i5-7200 CPU and 8 GB of RAM. All the initial parameters of the base classifiers adopt default values, and the value of training rounds is set to 50. The hyperparameters of MFGBoost and W-OD are selected according to different datasets.

5.2.1 Classification performance of benchmark imbalanced datasets

To further evaluate the performance of MFGBoost, four imbalance datasets (MFCCs, Page Blocks, AReM and MINIST) are utilized for multi-label classification. To illustrate the effectiveness of the improved focal loss of MFGBoost, we compare MFGBoost algorithm (nested with XGBoost or LightGBM) with the ordinary XGBoost and LightGBM algorithms and adopt the accuracy, macro-averaged f1-score (MAF1) and micro-averaged f1-score (MIF1) as the evaluation indexes. The classification results are shown in Tables 2 and 3.

In Table 2, MAF1 and MIF1 are the key indicators of multi-label imbalance classification, and MFGBoost performs better than the ordinary boosting model in all datasets. For Page Blocks dataset with the highest imbalance rate, MAF1 and MIF1 scores are far higher than those

Table 2 Classification performance of models on benchmark imbalanced datasets, in which the classification difficulty of different categories of samples is different

	Accuracy	MAF1	MIF1	Accuracy	MAF1	MIF1
	MFCCs ($\gamma = 2, \alpha = 1$)			Page Blocks ($\gamma = 2, \alpha = 1$)		
XGBoost	0.986	0.957	0.986	0.954	0.774	0.954
MFGBoost-XGBoost	0.984	0.976	0.984	0.982	0.89	0.982
	MFCCs ($\gamma = 2, \alpha = 1$)			Page Blocks ($\gamma = 2, \alpha = 1$)		
LightGBM	0.983	0.954	0.982	0.964	0.758	0.964
MFGBoost-LightGBM	0.987	0.988	0.987	0.978	0.839	0.978

Table 3 Classification performance of models on benchmark imbalanced datasets, in which the number of samples of different categories is different

	Accuracy	MAF1	MIF1	Accuracy	MAF1	MIF1
	AReM ($\gamma = 0, \alpha = 0.5$)			MINIST ($\gamma = 0.1, \alpha = 0.25$)		
XGBoost	0.793	0.805	0.793	0.974	0.973	0.974
MFGBoost-XGBoost	0.795	0.808	0.795	0.974	0.974	0.974
	AReM ($\gamma = 0, \alpha = 0.25$)			MINIST ($\gamma = 0.2, \alpha = 0.25$)		
LightGBM	0.789	0.802	0.789	0.969	0.969	0.969
MFGBoost-LightGBM	0.796	0.808	0.796	0.973	0.973	0.973

ordinary models, showing that MFGBoost model could learn each category better. The accuracy of MFGBoost is greatly improved, which shows that the improvement of minority categories' learning ability enhances overall accuracy when classifying extremely imbalanced datasets. The accuracy score of XGBoost is slightly higher than MFGBoost because XGBoost neglects the imbalance of the number of samples, which leads to the low classification accuracy of minority samples.

In Table 3, AReM and MINIST with the low imbalance rates include samples that are difficult to distinguish. MFGBoost could learn each category well when γ and α are less than 1, which could improve the ordinary model's classification index score.

5.2.2 Classification performance of sensors datasets

Two sensor datasets are selected to evaluate the denoise performance of W-OD in experiments. LightGBM is adopted as the base classifier W-OD. The human motion dataset and HAPT dataset include inertial sensor data. For HAPT dataset, we adopt W-OD to denoise the inertial sensor data and adopt the window matrix designed in this paper to sample the data. The denoising and sampling method for building HAPT dataset is compared with our method in Table 4. thr represents the threshold of noise influence degree. When the output of COPOD is more than thr, the data is greatly affected by noise. In this case, ξ_1 is taken as the loss function weights. On the contrary, ξ_2 is taken as the loss function weights when the output of COPOD is lower than the thr.

In Table 4, the accuracy of W-OD with di fferent parameters is higher than ordinary model. When $\xi_1 = 3$ and $\xi_2 = 0.8$, W-OD is with the highest accuracy, which shows the best performance.

For the human motion dataset, the classification accuracy comparison between W-OD-LightGBM and W-LightGBM without denoise is shown in Table 5.

Further experiments on the human motion dataset are given in Section 5.3.

Table 4 The classification performance on HAPT dataset based on different W-OD parameters

	thr	ξ_1	ξ_2	Accuracy
LightGBM	–	–	–	0.907
W-OD-LightGBM-1	5	2	1	0.928
W-OD-LightGBM-2	10	2	1	0.933
W-OD-LightGBM-3	15	2	1	0.935
W-OD-LightGBM-4	15	3	1	0.935
W-OD-LightGBM-5	15	3	0.8	0.939

5.2.3 The ablation studies and contrast experiment of W-OD-MFGBoost

In the ablation experiment and contrast experiment, we choose the conservative values, $\beta_1 = 1$, $\beta_2 = 2$, $\varepsilon_1 = 2$, $\varepsilon_2 = 1$ and $thr = 15$. The parameters of W-OD-LightGBM and MFGBoost-LightGBM are changed accordingly.

Taking LightGBM as the base classifier, we discuss the influence of a single component of W-OD-MFGBoost on classification performance from four aspects, Accuracy, Macro-averaged precision (MAP), Macro-averaged recall (MAR) and MAF1. MFA1 and MAR are the key indicators of multi-label imbalance classification, and MAP and Accuracy represent the model's classification accuracy under the influence of noise. This ablation experiment is carried out on HAPT dataset with a certain imbalance rate and signal-to-noise ratio. The inertial sensor signals of HAPT dataset are sampled and filtered according to the method

Table 5 Classification performance on human motion dataset

	thr	ξ_1	ξ_2	Accuracy
W-LightGBM	-	-	-	0.965
W-OD-LightGBM	20	2	1	0.973

Table 6 The ablation studies of W-OD-MFGBoost

	Accuracy	MAP	MAR	MAF1
LightGBM	0.907	0.771	0.776	0.767
W-OD-LightGBM	0.935	0.82	0.79	0.799
MFGBoost-LightGBM	0.915	0.83	0.83	0.825
W-OD-MFGBoost-LightGBM	0.933	0.82	0.772	0.792

in [46] to form the normal samples, so the original inertial sensor data and normal samples are used in comparison between W-OD-LightGBM and LightGBM.

In Table 6, W-OD-LightGBM has a significant improvement in accuracy and MAP compared with LightGBM, which shows that W-OD algorithm improves the model's anti-interference ability and classification accuracy. In terms of MAR and MAF1, MFGBoost-LightGBM has a significant improvement compared with the LightGBM, which shows that MFGBoost enhances the performance of classifying imbalanced data. W-OD-MFGBoost-LightGBM is formed by combining the W-OD algorithm and MFGBoost-LightGBM model, which has an excellent performance in all four indicators.

Figure 10 shows the influence of the values of the parameters on focal loss. For α taking a smaller value and γ taking a larger value, the imbalanced data classification ability of the model could be improved. However, for noise data, smaller α and larger γ could decrease its gradient relative ordinary data, which reduces the model's attention to noisy data. W-OD-MFGBoost balances the ability of denoise and label-imbalance classification. Although the performance on a single index is not as good as W-OD-LightGBM or MFGBoost-LightGBM, W-OD-MFGBoost-LightGBM has a balanced performance in four indicators, and its performance is better than original LightGBM model.

To verify the superiority of our proposed model in multi-label classification, several newly models are compared and carried out experiments on HAR dataset. LightGBM, CATBoost [16] and NGBoost [48] are the most popular gradient boosting learning algorithms, and they have outstanding performance in multi-label classification tasks.

Table 7 Comparison between W-OD-MFGBoost and other latest classification models

	Accuracy	MAP	MAR	MAF1
PGBM [49]	0.892	0.90	0.887	0.889
NGBoost [48]	0.881	0.884	0.875	0.876
CATBoost [16]	0.899	0.902	0.895	0.897
AMF [51]	0.905	0.907	0.899	0.901
PRF [50]	0.927	0.927	0.924	0.925
LightGBM [4]	0.926	0.928	0.924	0.925
W-OD-MFGBoost-LightGBM	0.965	0.965	0.964	0.964

PGBM [49] is a new probabilistic gradient boosting framework. Probabilistic Random Forest (PRF) [50] and Aggregated Mondrian forests for online learning (AMF) [51] are the extensions of random forest algorithms, representing the latest progress in the research of random forest algorithms. In particular, PRF is suitable for classifying noisy datasets, so PRF is compared with W-OD-MFGBoost-LightGBM to verify the reliability performance of W-OD algorithm. The results are shown in Table 7.

In Table 7, W-OD-MFGBoost-LightGBM has higher accuracy and MAP than other algorithms, and its performance is better than other algorithms in terms of imbalance index.

Figure 12 shows the ROC curve of each classifier, and Fig. 13 shows the PR curve of each classifier. W-OD-MFGBoost has better classification performance than other models.

5.3 Model experimental analysis of the intelligent walking stick

The laptop is utilized to carry out the experimental analysis of the models. In MFGBoost and W-OD algorithm training, Tree-structured Parzen Estimator (TPE) algorithm is adopted to search the optimal hyperparameters. The new W-OD-MFGBoost-LightGBM algorithm is transplanted into the Raspberry Pi zero w to realize the classification of walking motions.

5.3.1 Hyperparameter search

In this section, Bayesian optimization is adopted to search for optimal hyperparameters. Hyperopt [52] is a Python library that provides algorithms and parallelization infrastructure for performing hyperparameter optimization.

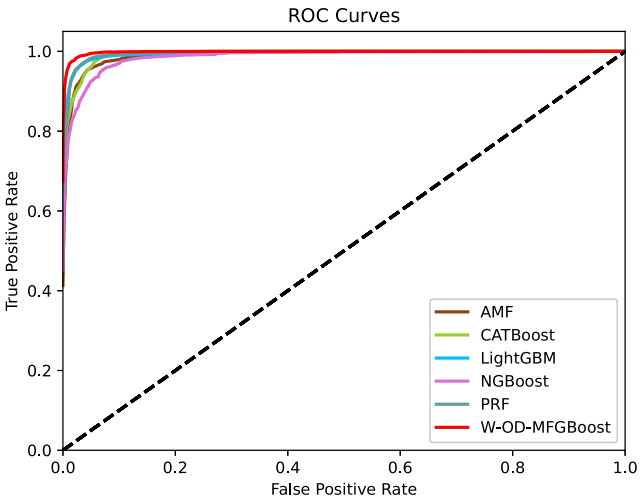


Fig. 12 ROC Curve of classifiers

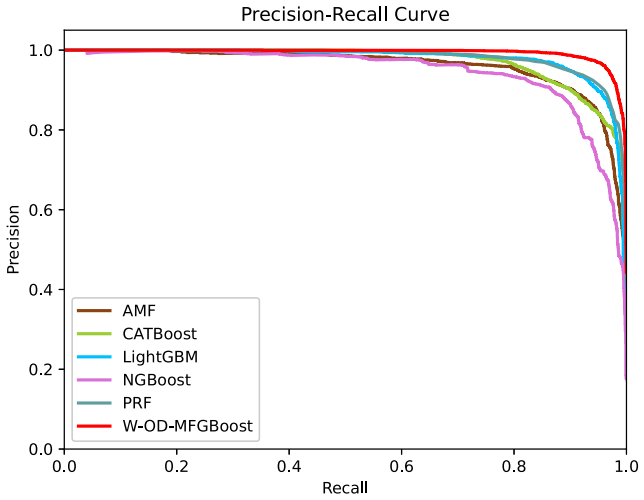


Fig. 13 Precision-Recall Curve of classifiers

Bayesian optimization is a method for optimizing the objective function. It is suitable for optimization in a continuous domain with less than 20 dimensions. TPE algorithm is widely used in hyperparameter optimization of the ensemble learning model due to its high efficiency. In this paper, TPE algorithm is built based on the Hyperopt library. W-OD-MFGBoost model is set to objective function of Hyperopt, and ACCN and MFA1 are taken as the optimization goal, which is shown in formula (22).

$$Optimization\ goal = ACCN + MAF1, \tag{22}$$

where $ACCN$ represents the accuracy rate of samples affected by noise greater than the threshold, and $MAF1$ represents the classification accuracy of the model.

The value intervals of hyperparameter β_1, β_2, ε_1, ε_2 are [0,10], [0,10], [0,1] and [0,1], respectively. Based on the data output by COPOD, the value of thr is set at 30. After 300 iterations, the optimization process based on TPE algorithm is shown in Fig. 14.

When β_1 taking smaller values and β_2 taking larger value, the optimization of the target value is better. The changing regularities of β_1, β_2 and ε_2 demonstrate the previous analysis in Section 4.1. ε_1 has little influence on the optimization goal, which may be related to the data value of the dataset.

In this case, the best values of hyperparameters are $\beta_1 = 2.2144054012685253$, $\beta_2 = 7.1896167625029435$, $\varepsilon_1 = 0.514618997993338$, $\varepsilon_2 = 0.5210469880025101$ and $thr = 30$.

5.3.2 Performance analysis

In this section, we verify the superiority of W-OD-MFGBoost-LightGBM model over other latest imbalanced data classification models. These models are trained on the human motion dataset. The number of training rounds is set at 50, and the rest of the parameters are selected as default values. Since the code of FLAGB is not open source, the custom loss function of LightGBM is utilized to build a binary FLAGB model according to the formula provided in [39], which is extended to multi-classification by OVR algorithm. There is no case of multi-classification in CUSBoost [34], RUSBoost [35] and Imbalance XGBoost [11]. In order to facilitate comparison, they are converted into multi-classification models through the OVR algorithm.

The comparison among the latest algorithms is shown in Table 8, which shows the recall rate of each category and accuracy. W-OD-MFGBoost-LightGBM's recall rate of each category is higher than other models. For the category that is difficult to classify, such as "slow walking" and "fast walking", the classification effect of W-OD-MFGBoost-LightGBM is much higher than those of other models. This shows that W-OD-MFGBoost-LightGBM has better performance in class imbalance. W-OD-MFGBoost-LightGBM achieved the highest score for accuracy, which shows that our model has the better performance in anti-interference performance.

In summary, W-OD algorithm effectively reduces the influence of noise in the training process. For the recall rates of each category, W-OD-MFGBoost-LightGBM performs

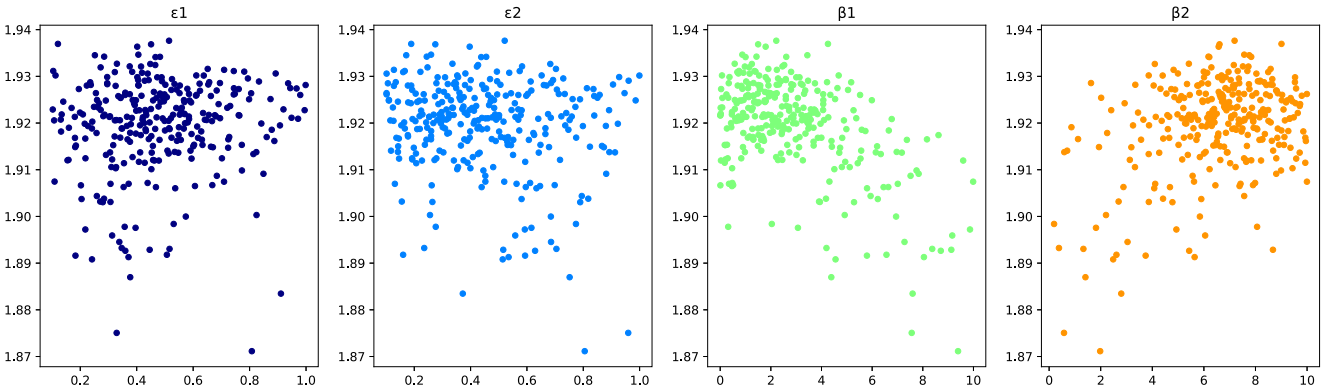


Fig. 14 The Bayesian optimization process

Table 8 Accuracy and category recall rate of different models

Classifiers	Accuracy	Slow walking	Fast walking	Turning right	Turning left
CUSBoost [34]	0.954	0.94	0.944	0.947	0.952
RUSBoost [35]	0.937	0.922	0.934	0.928	0.935
FLAGB [39]	0.949	0.931	0.941	0.937	0.946
Imbalance XGBoost [11]	0.969	0.96	0.957	0.963	0.969
NGBoost [48]	0.921	0.893	0.921	0.905	0.917
CATBoost [16]	0.959	0.949	0.953	0.952	0.957
PGBM [49]	0.842	0.834	0.874	0.855	0.852
AMF [51]	0.918	0.902	0.914	0.919	0.929
PRF [50]	0.962	0.954	0.961	0.959	0.963
LightGBM [4]	0.968	0.963	0.961	0.965	0.97
W-OD-MFGBoost-LightGBM	0.97	0.977	0.968	0.969	0.971

best, which shows that our improvement strategy has good performance for dealing with class imbalances. Thus, the new W-OD-MFGBoost-LightGBM model is more effective than the existing models.

5.4 Applications and development

MFGBoost is highly scalable because it primarily improves and solves loss functions rather than making deep customizations. MFGBoost not only suits for traditional boosting algorithms but also provides ideas for deep learning algorithms to solve class imbalance problems. The deep boosting models were proposed in [53–55], which used the convolution neural network (CNN) as the basic learner of the boosting algorithm. The principle of MFGBoost may be applied to deep boosting models to deal with class imbalances. The neuromorphic computing is investigated in [56–59], which reduces the amount of calculation and power consumption. Due to existence of imbalanced samples, it is necessary to study class imbalance on SNN based on multi-classification focal loss.

W-OD is a general scheme to solve the influence of noise on classification results, which has no requirement for classifier and loss function. Due to the advantages of COPOD algorithm, W-OD can deal with sensor noise and reduce the impact of other types of outliers.

6 Conclusion

In this paper, a new walking motion real-time detection method is designed, which is based on the walking stick, IoT, COPOD, and improved LightGBM. Raspberry Pi and MPU-6050 were affordable and effective in data acquisition and real-time detection of walking motions. Based on acceleration and gyroscope sensors, several kinds of walking motion data are collected. The new MFGBoost and W-OD are proposed in this paper, which shows excellent performance in walking stick motion classification and strong anti-interference performance. To the best of our knowledge, MFGBoost and W-OD-MFGBoost are the first of their kind, which provides a general extension of the boosting model for multi-label classification imbalance scenarios and brings a weighted denoising method without changing data. They are embedded into Raspberry Pi to realize real-time detection. IoT technology is designed to transmit real-time classification results and analyze motion behavior. The motion behaviors are monitored in real-time through the designed APP and IoT system.

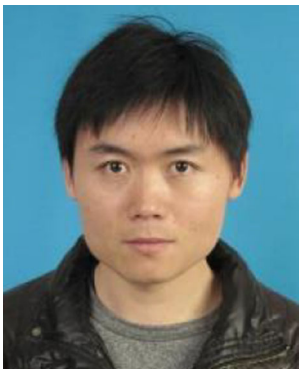
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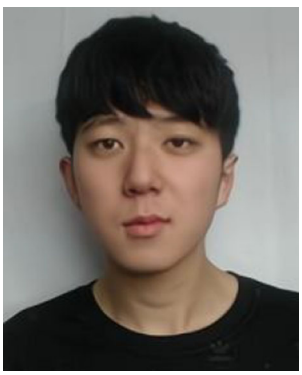
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